AgroNex-Pi: Revised Physical Pin Configuration

Complete Interconnect Map • Raspberry Pi 4B • Arduino Uno Slave • ADS1115 ADC • Relays & Actuators

1. Hardware Architecture Overview

The **AgroNex-Pi** robot employs a master-slave embedded computing architecture. The **Raspberry Pi 4 Model B** serves as the primary compute unit, running the Linux OS, OpenCV computer vision, TensorFlow Lite deep learning inference, SQLite telemetry storage, and AWS Cloud Gateway relay. Analog instrumentation (Capacitive Soil Probe v1.2 and HW-038 Water Reservoir Sensor) is digitized via a precision **ADS1115 16-Bit I2C ADC** (Address 0x48). Low-level motor PWM timing, ultrasonic obstacle distance radars, and the **3D-printed Soil Moisture Probe Linear Actuator** are managed by an **Arduino Uno R3** coprocessor connected over USB Serial. The opto-isolated 5V relay module controls the 12V submersible irrigation pump and grass trimmer cutter with **independent, simultaneous switching**.

Key Operational Constraints & Chronology:
• ***Startup Sequence:*** *Boot -> IDLE (3s) -> Actuator Probe Moves DOWN -> All Sensors Sampled -> Actuator Probe Retracts UP -> Chassis Drives FORWARD -> Autonomous Patrol.*
• ***Subsystem Periodic Timing:*** *Soil Moisture Actuator probes every **1 Hour** • Ambient Temperature scans every **10 Minutes** • HW-038 Water Level monitors **Continuously (Zero Delay)**.*
• ***Independent Relay Channels:*** *Water Pump (CH1) and Grass Cutter (CH2) are decoupled and can operate concurrently without mutual rejection.*

2. Raspberry Pi 40-Pin GPIO Header Mapping (BCM Specification)

Physical Pin	BCM GPIO	Signal Type	Connected Device / Subsystem	Wiring Details & Logic Level
Pin 1	3.3V Power	DC Power	ADS1115 VDD, DHT VCC, HMC5883L	Regulated 3.3V supply from Pi PMIC (<50mA load)
Pin 2	5.0V Power	DC Power	Relay Module VCC, Buzzer VCC	5V logic supply rail (backed by 3A LM2596 buck converter)
Pin 3	GPIO 2	I2C1 SDA	ADS1115 ADC & HMC5883L Compass	Hardware I2C Data line (internal 1.8k pull-up to 3.3V)
Pin 5	GPIO 3	I2C1 SCL	ADS1115 ADC & HMC5883L Compass	Hardware I2C Clock line (100 kHz bus clock)
Pin 6, 9, 14	GND	Ground	Common System Star Ground	Unified GND plane meeting Pi, battery, & sensors
Pin 7	GPIO 4	Digital 1-Wire	DHT11 / DHT22 Sensor DATA	Atmospheric temp & humidity; 10k pull-up to 3.3V
Pin 10	GPIO 15	UART RXD	NEO-7M GPS Module TXD	9600 Baud NMEA serial sentences stream
Pin 11	GPIO 17	Digital OUT	Relay CH1: Submersible Water Pump	Active-LOW (0V = Energized/Pumping, 3.3V = De-energized)
Pin 13	GPIO 27	Digital OUT	Relay CH2: Grass Cutter Motor	Active-LOW (0V = Energized/Trimming, 3.3V = De-energized)
Pin 15	GPIO 22	Digital OUT	Active 5V Alarm Buzzer	Driven via 2N2222 NPN transistor buffer for loud alerts
Pin 16	GPIO 23	Digital OUT	System Heartbeat / Fault LED	Red LED Anode via 220 Ohm current-limiting resistor
Pin 29, 31	GPIO 5, 6	PWM / GPIO	BTS7960 #1 RPWM / LPWM (Left 4WD)	Chassis Left Motor control (Pi direct fallback port)
Pin 33, 35	GPIO 13, 19	PWM / GPIO	BTS7960 #2 RPWM / LPWM (Right 4WD)	Chassis Right Motor control (Pi direct fallback port)
USB 3.0 Port	/dev/ttyACM0	USB CDC ACM	Arduino Uno Slave Controller	115200 Baud link for Actuator Probe & Motor commands
CSI Ribbon	CSI-2 Bus	2-Lane MIPI	Raspberry Pi CSI Camera Module	Captures real-time leaf foliage frames for CNN inference

3. ADS1115 16-Bit I2C ADC Channel Map

To avoid noise on the Pi's internal SPI bus and eliminate ground shifts, all analog field sensors connect to the dedicated **ADS1115 I2C Analog-to-Digital Converter** (Operating Address: 0x48).

Channel	Connected Sensor	Operating Principle	Raw ADC Range (16-Bit)	Calibrated Engineering Metric
AIN0	Capacitive Soil Moisture Probe v1.2	Dielectric capacitance frequency shift; corrosion-resistant	Dry: 21000 Submerged: 9500	0.0% to 100.0% Volumetric Water Content (VWC)
AIN1	HW-038 Water Reservoir Level Sensor	Parallel conductor resistance immersion depth; 3-pin	Empty: 1200 Full (4cm): 18500	0.0% to 100.0% Reservoir Capacity (Continuous)
AIN2	Battery Voltage Divider (12V Pack)	10k/2.2k precision divider (5.5:1 ratio into 3.3V range)	9.0V: 15400 12.6V: 21560	9.0V to 12.6V Li-ion Battery Pack Voltage
AIN3	Spare Analog Channel	Reserved for optional solar panel monitoring or pH probe	0 to 32767	General Analog Input (0 - 4.096V range)

4. Arduino Uno Slave Controller (Motors, Actuator & Radars)

The Arduino Uno receives serial string commands from the Pi over USB (/dev/ttyACM0 at 115200 baud). It generates stable hardware PWM for the BTS7960 motor drivers and servo pulses for the **3D-Printed Soil Probe Linear Actuator**.

Arduino Pin	Subsystem Component	Signal Direction	Hardware Role & Command Protocol
Pin D3 (PWM)	Soil Moisture Actuator Servo	Digital OUT (50 Hz PWM)	**PROBE_DOWN:** Rotates arm into soil. **PROBE_UP:** Lifts arm safely back to travel posture.
Pin D4	Left BTS7960 Enable (R_EN + L_EN)	Digital OUT (5V)	Bridges R_EN and L_EN of Left BTS7960 driver to enable H-bridge.
Pin D5, D6	Left BTS7960 (Left 4WD Motors)	PWM Output (Timer 0)	**D5:** Left Forward PWM (RPWM). **D6:** Left Reverse PWM (LPWM). Driven via 'M,,'.
Pin D7	Right BTS7960 Enable (R_EN + L_EN)	Digital OUT (5V)	Bridges R_EN and L_EN of Right BTS7960 driver to enable H-bridge.
Pin D9, D10	Right BTS7960 (Right 4WD Motors)	PWM Output (Timer 1)	**D9:** Right Forward PWM (RPWM). **D10:** Right Reverse PWM (LPWM).
Pin D2, D8	Front Ultrasonic Sensor (HC-SR04)	D2: TRIG (OUT) D8: ECHO (IN)	Measures forward obstacles. Commanded via 'SCAN_F,'. Reports 'DIST_F:'.
Pin D11 (PWM)	Front Ultrasonic Pan Servo	Digital OUT (50 Hz PWM)	Sweeps front radar bracket left/right during obstacle avoidance.
Pin D12 (PWM)	Rear Ultrasonic Pan Servo	Digital OUT (50 Hz PWM)	Sweeps rear radar bracket left/right during reverse patrol.
Pin A0, A1	Rear Ultrasonic Sensor (HC-SR04)	A0: TRIG (OUT) A1: ECHO (IN)	Measures rear obstacle clearance. Reports 'DIST_R:'.
USB Type-B	Raspberry Pi 4 USB 3.0 Port	Bidirectional Serial (115200)	Full-duplex command link. Auto-halts motors if communication watchdog drops >500ms.

5. Opto-Isolated Relay Subsystem & Electrical Decoupling

Independent Simultaneous Switching: CH1 (Irrigation Pump) and CH2 (Grass Cutter) operate on separate opto-isolated relay channels driven by GPIO 17 and GPIO 27. Each channel features an onboard PC817 optocoupler providing 2.5 kV isolation between the Raspberry Pi's digital 3.3V logic and the high-current 12V DC motor circuits. Flyback diodes (1N4007) are placed in reverse parallel across both the water pump and cutting motor terminals to suppress inductive kickback spikes (Back-EMF) that would otherwise cause voltage sags or brownouts on the Pi.

6. Operational Startup Chronology & Autonomous Scheduling

Phase / Event	Trigger Mechanism	Hardware Motion & Sensor Action	Dashboard Telemetry Result
Step 1: IDLE	System Boot / Reset	Robot remains stationary for 3.0 seconds to allow sensor warm-up & GPS fix.	Status: 'IDLE'; Connection dot glows green.
Step 2: Probe Down	Startup Chronology	Linear actuator extends probe DOWN into soil (PROBE_DOWN pulse).	Actuator pill: 'PROBING (DOWN)'.
Step 3: Sample Sensors	Startup Chronology	Reads Capacitive Soil Moisture, HW-038 Water Level, DHT Temp/Hum, GPS, Compass.	Live metrics update with persistent values.
Step 4: Probe Up	Startup Chronology	Linear actuator retracts probe safely UP into travel position (PROBE_UP).	Actuator pill: 'RETRACTED (UP)'.
Step 5: Move Forward	Startup Chronology	Chassis dual BTS7960 motors drive FORWARD for 2.0s at 60% duty cycle.	FSM: 'NAVIGATING' -> 'MONITORING'.
Periodic 1-Hr Actuator	Every 3600 seconds	Halts chassis, lowers probe, samples moisture, retracts probe, updates database.	Last moisture reading persistently displayed.
Periodic 10-Min Temp	Every 600 seconds	DHT sensor reads ambient temperature & humidity, checking crop heat stress.	Last temperature reading persistently displayed.
Continuous Water Level	Real-Time (Zero Delay)	HW-038 sensor analog read every 1s, preventing water tank dry-running.	Instant reservoir gauge update on dashboard.